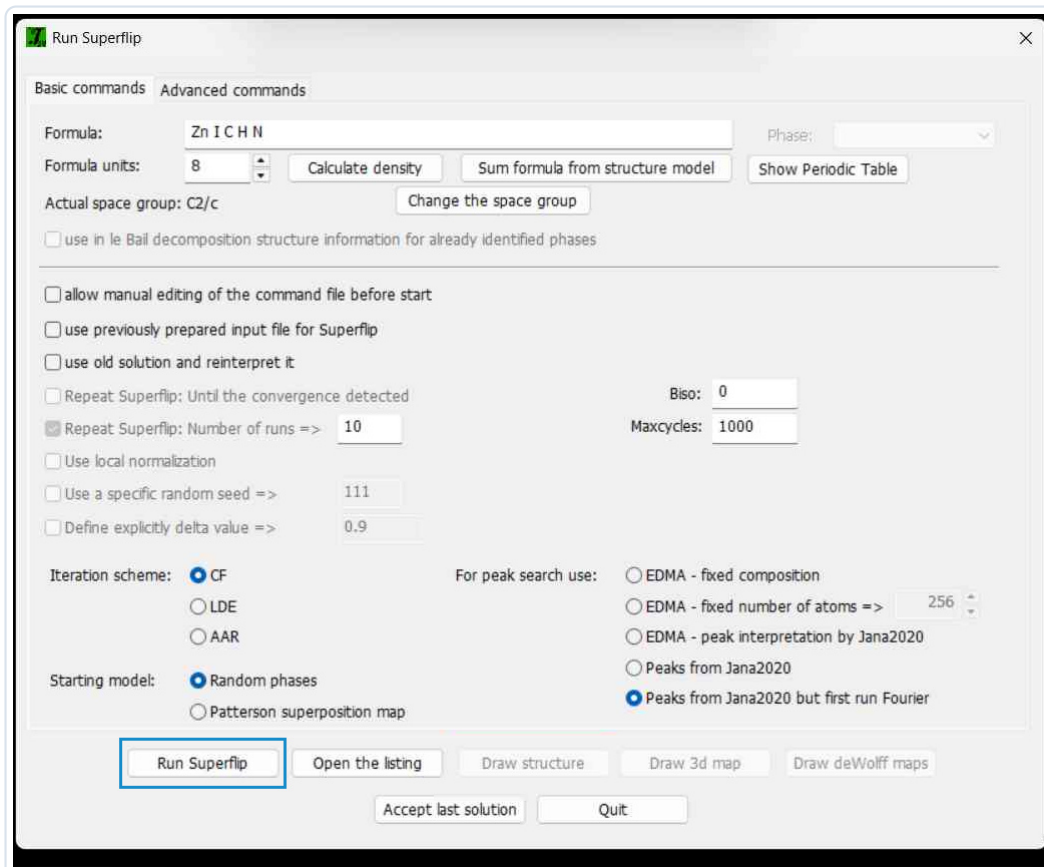


First use and setup

Start Phase Studio from Jana2020 and configure the first workflow

This guide assumes that the Phase Studio Jana2020 integration is already installed and that the reflection data have already been prepared and indexed in Jana2020.

- Start Phase Studio from Jana2020**
In Jana2020, open the Run Superflip dialog. After the reflections are ready, click Run Superflip. With the Phase Studio integration installed, this launches Phase Studio for the current Jana2020 dataset.

**NOTE**

Phase Studio is launched from the standard Jana2020 Superflip workflow. You do not need to start a separate desktop application.

Configure Phase Recycling

Choose the recycling options for your first run

2

Configure Phase Recycling

Choose the map used for phase recycling, set the number of recycling cycles and select the SharpED model. Then click Next.

The screenshot shows the 'PHASE RECYCLING' configuration window. The title bar reads 'Superflip - SharpED - EDVIA workflow'. The main header is 'PHASE RECYCLING' with the subtitle 'Configure phase recycling and validation'. The input field shows 'Jana2020 input: resol_2cela5_auto_Superflip.inflip'. Below this, a label reads 'Map used for phase recycling and Jana2020 handoff'. There are two radio buttons: 'SharpED map' (unselected) and 'Superflip map' (selected). The 'Recycling cycles' is a numeric input set to '3'. The 'Model' is a dropdown menu set to 'koala 4.0'. Below the model dropdown is a 'Refresh models' button and the text '11 models available · server default: koala 4.0'. A 'SharpED settings' button is located below the model selection. The 'Cross-validation' section contains two checkboxes: 'Compute OMIT validation maps' and 'Calculate R_free', both of which are unchecked. A descriptive text below the checkboxes states: 'Optional. Excludes a random 5% of reflections each cycle to compute OMIT / R_free validation metrics, which help rank the recycling cycles and select the most suitable map.' At the bottom, there are four buttons: '< Back', 'Cancel', 'Open full configuration', and 'Next >'. A mouse cursor is hovering over the 'Next >' button.

Review settings and connect SharpED

Continue to the Map Feedback page and open SharpED account access

3

Review Map Feedback settings

Keep or adjust the available feedback options as needed. For a first run, the default settings can usually be left unchanged. Click Run phasing.

PHASE RECYCLING · MAP FEEDBACK
Optionally update reflection data between recycling cycles

Jana2020 input: resol_2cela5_auto_Superflip.inflip

Reflection data

Type Single crystal

Format h k l · F · phase · $\sigma(F)$

Available feedback methods for single-crystal reflection data.

Warning
The operations on this page modify the reflection data supplied to subsequent cycles. Results from these cycles should therefore be validated against the original measured data.

Missing-reflection completion

☐ Enable missing-reflection completion

Start after cycle 1

Maximum added reflections (%) 0.000

Intensity correction

☐ Enable intensity correction

Start after cycle 1

Correction damping 0.000

< Back Cancel Open full configuration Run phasing

4

Open SharpED account access

If no API token is configured yet, Phase Studio shows the SharpED Access Required dialog. Click Open SharpED account.

Phase Studio — SharpED access required

PHASE STUDIO 1.0.9
Superflip · SharpED · EDMA workflow

SHARPEd ACCESS REQUIRED
Configure access before SharpED processing starts

No SharpED API token is configured.
SharpED processing requires an active Jana2020 or SharpED account and an API token.

API token Enter API token

Open SharpED account Set token Skip for now

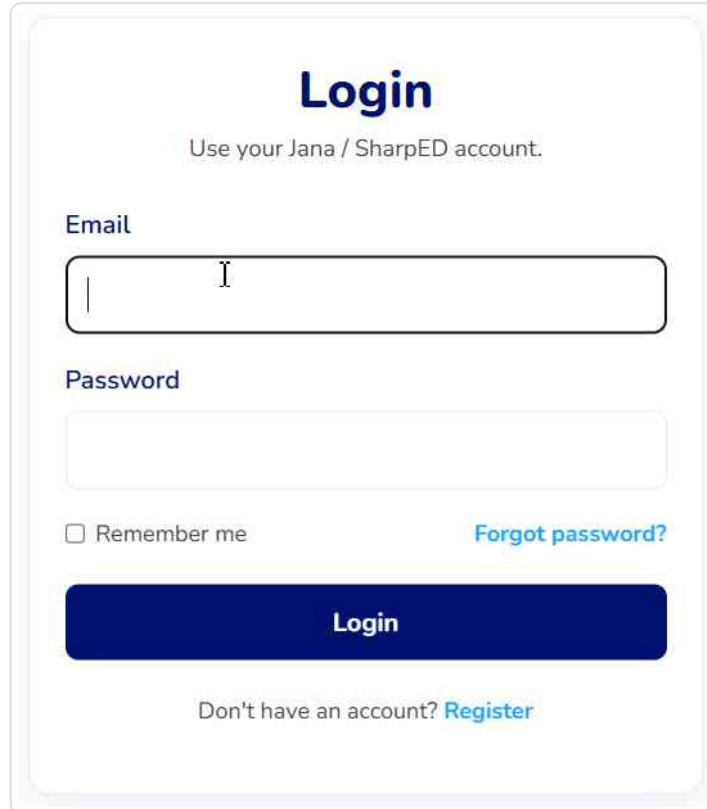
Open the website and create a token

Use the SharpED web page to sign in and access the token controls

5

Sign in if needed

The browser opens the SharpED website. Sign in with your Jana or SharpED account. The login fields are intentionally shown without credentials in this guide.



The image shows a login form titled "Login" with the subtitle "Use your Jana / SharpED account." It contains two input fields: "Email" and "Password". Below the password field is a checkbox labeled "Remember me" and a link "Forgot password?". A large blue "Login" button is centered below these fields. At the bottom, there is a link "Don't have an account? Register".

TOKEN PAGE <https://sharped.fzu.cz/sharp-ed>

6

Open the API token section

On the SharpED Tool page, expand API tokens for desktop application by clicking Show.

API tokens for desktop application

[Show](#)

API tokens allow your C++ application to submit jobs to the queue directly, without logging in through a browser. Use <https://sharped.fzu.cz> as the only API host (web, Phase Studio, and inference).

7

Create a token

Enter a token name that identifies this computer or installation, then click Create token.

API tokens for desktop application

[Show](#)

API tokens allow your C++ application to submit jobs to the queue directly, without logging in through a browser. Use <https://sharped.fzu.cz> as the only API host (web, Phase Studio, and inference).

New token name

[+ Create token](#)

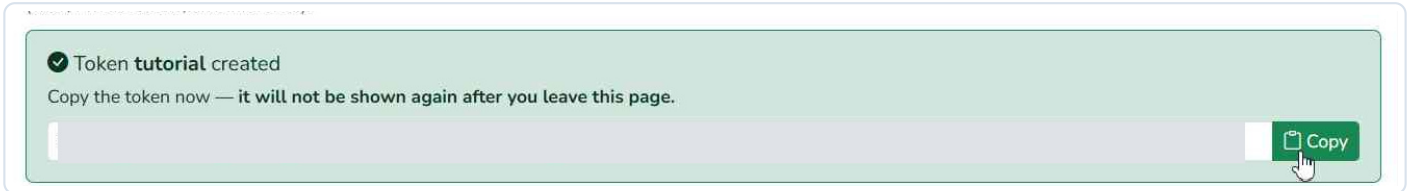
Copy the token and return to Phase Studio

Only the token value itself is hidden below; the rest of the interface remains visible

8

Copy the generated token

Copy the generated token immediately. The actual token value is covered in the screenshot below for security reasons.



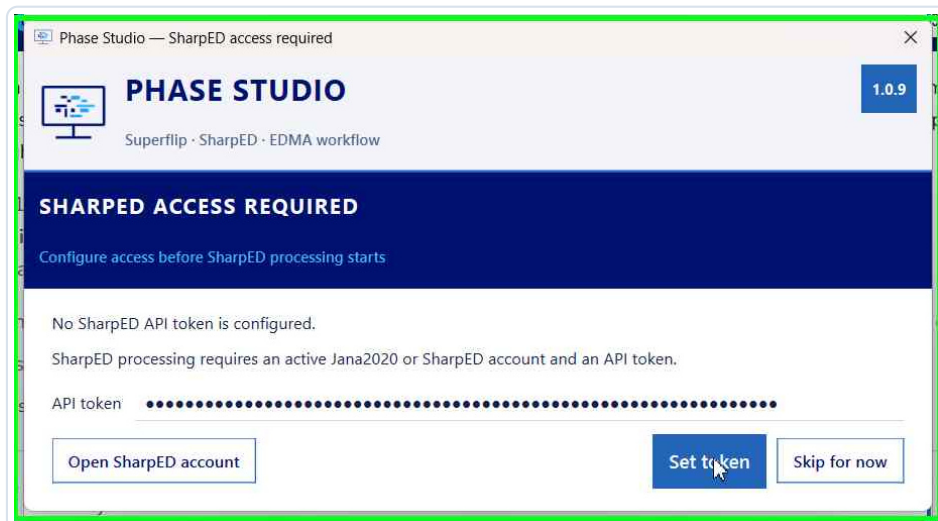
SECURITY

The generated API token is shown only once on the website. Copy it immediately and do not share it or include it in screenshots or public reports.

9

Paste the token into Phase Studio

Return to Phase Studio, paste the copied value into the API token field and click Set token. The screenshot shows only masked bullet characters, not the token itself.



Let the workflow run

Monitor the status, metrics and structure comparison while processing is in progress

10

Let Phase Studio finish the recycling cycles

After the token is accepted, the workflow continues. The run status, metrics and structure comparison update while processing is in progress.

The screenshot displays the Phase Studio interface during a workflow run. The interface is divided into several sections:

- Data input:** Includes fields for Input mode (Jana2020 .inflip), Jana2020 .inflip file, External HKL file, and HKL format (hkl F phase sigma). Buttons for 'Validate HKL' and 'Analyze completeness' are present.
- Crystal metadata:** Displays metadata source (Jana2020 .inflip) and cell parameters: Cell lengths (30.90300 × 15.17560 × 35.32600 Å), Cell angles (α 90.000° × β 102.560° × γ 90.000°), Space group (C2/c (#15)), and Composition (Zn8 I8 C8 H8 N8).
- Running...:** A section with buttons for 'Continue run', 'Pass to Jana2020', 'Stop after current cycle', 'Stop immediately', and 'Clear results'.
- RUN STATUS:** Shows the overall status as 'Running' and the current cycle as 'Cycle 3 of 3 - SharpED - Processing'. A progress bar indicates the stage (Stage 4 of 6). A note states 'Configuration locked while the calculation is running.'
- WORKFLOW METRICS:** Displays 'Assessment: Reference-free · no holdout'. A graph shows 'Amplitude R.F.' (0.0 to 0.4) versus 'Cycle' (1 to 3). The graph compares 'Superflip' (blue line) and 'SharpED' (red line). The 'Amplitude R.F.' for Superflip is approximately 0.4, while for SharpED it is approximately 0.0.
- STRUCTURE COMPARISON:** Shows three views: 'Reference' (No reference structure selected), 'Superflip' (748 atoms shown), and 'SharpED' (656 atoms shown). A note indicates 'Drag to rotate · right-drag to zoom · middle-drag to pan — all views stay in sync · Hydrogen and helium atoms hidden'.
- EXECUTION LOG:** Displays the output file 'cycle_003\edma_superflip\cycle_003_superflip_edma.cif', the SharpED server HTTP timeout (600 seconds), and the upload and job submission details for 'cycle_003_superflip.xplor' (27.1 MiB, ID 69).

Return the result to Jana2020

Select the result and confirm the final information message

11

Select the result and pass it to Jana2020

When the workflow is complete, review the result list. Keep the desired result selected and click Pass to Jana2020.

JANA2020 RESULT SELECTION
Compare completed maps and select the result to pass to Jana2020

Assessment: **Reference-free · no holdout**
Completed cycles: **3**
Recommended result: **Cycle 1 · Superflip**
Reference: **Not available**
Holdout: **Not enabled**

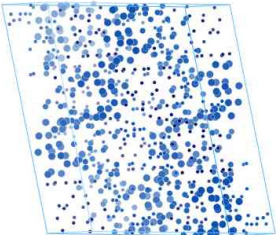
Recommended by lowest Amplitude R_F; Amplitude CC and Weighted triplet C3 are tie-breakers.

RESULT CANDIDATES

Recommended	Cycle	Source	Amplitude R_F ↓	Amplitude CC ↑	Weighted triplet C3
Recommended	1	Superflip	0.000	1.000	0.539
	1	SharpED	0.442	0.797	0.311
	2	Superflip	0.000	1.000	0.523
	2	SharpED	0.441	0.798	0.339
	3	Superflip	0.000	1.000	0.523
	3	SharpED	0.440	0.799	0.331

STRUCTURE PREVIEW

Cycle 1 · Superflip



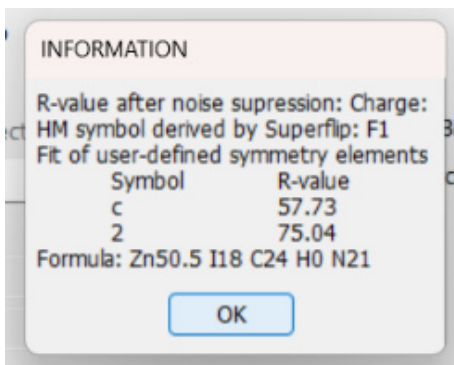
788 atoms shown

[Return to Phase Studio](#) [Pass to Jana2020](#)

12

Confirm the Jana2020 information message

After the result is transferred, Jana2020 can show an information dialog with the R-value after noise suppression and the fitted symmetry elements. Review it and click OK.



Done - SharpED access is configured and the first Phase Studio workflow is complete.

The selected result has been returned to Jana2020 for the next steps of the structure solution workflow.